

Interaction of Heat and Pollution and its impact on *Jhuggi Jhopris* of New Delhi

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Abstract

When megacities expand, two environmental stressors intensify: urban heat and air pollution. While traditionally studied as parallel phenomena, aerosols and trace gases may act as a thermal blanket, trapping longwave radiation and inhibiting nocturnal urban cooling. This paper investigates the spatial inequality of this mechanism, examining whether the compounding effects of heat and pollution disproportionately burden morphologically vulnerable neighborhoods. Focusing on New Delhi, an important case study for rapidly urbanizing regions in the Global South, we constructed a high-resolution spatial panel ($N = 11,212$) over three pre-monsoon summers (April–June, 2022–2024). Harmonizing MODIS Land Surface Temperature and Sentinel-5P pollution data (AOD and NO_2) with administrative coordinates from the Delhi Urban Shelter Improvement Board (DUSIB), we employed a Two-Way Fixed Effects (TWFE) panel regression. Our results reveal a strong city-wide association: a one-unit increase in Aerosol Optical Depth is associated with a 1.02°C reduction in the nocturnal cooling rate.

Furthermore, grid cells containing recognized informal settlements show an additional 0.20°C reduction during high-AOD events compared to formal-area cells ($p < 0.01$). We emphasize that, given the observational design, these findings establish robust associations rather than causal effects, and we discuss the assumptions required for a causal interpretation along with their limitations. The results suggest that urban form shapes climate vulnerability in ways that demand hyper-localized policy responses.

Significance Statement: Urban heat islands and air pollution are primary public health threats in the Global South, yet their interactive effects are rarely mapped to localized urban inequality. This study provides empirical evidence that daytime smog is strongly correlated with reduced nighttime cooling, and that this association is more severe within formally recognized informal settlements. By linking satellite meteorology with official administrative slum coordinates in New Delhi, we find that residents of DUSIB-designated JJ clusters bear extra climate burden. This research underscores the need to bridge atmospheric science and environmental justice, and suggests that adaptation policies must account for the unique structural vulnerabilities of informal urban morphology.

1. Introduction

The simultaneous expansion of concrete infrastructure and severe degradation of air quality characterize rapid urbanization in megacities of the Global South. While the Urban Heat Island (UHI) and Urban Pollution Island (UPI) effects are well-documented as separate phenomena, emerging atmospheric research indicates they interact. Cao et al. (2016) proposed that daytime aerosols can act as a thermal blanket, trapping outbound longwave radiation and preventing urban canopies from cooling at night. Siddiqui et al. (2024) and Singh et al. (2024) have since observed spatial correlations between UHI and UPI intensity across Indian megacities. However, their analyses operate at city-wide aggregate scales, masking potentially severe micro-geographic inequalities.

Urban form fundamentally alters how heat and pollution interact at the neighborhood level. Informal settlements, characterized by extreme structural density, galvanized iron roofing, and negligible ventilation corridors, are morphologically predisposed to trap both heat and locally generated emissions. Building on Bhan et al.'s (2020) spatial assessments of housing inequality in New Delhi, this paper integrates official Jhuggi Jhopri (JJ) cluster boundaries into a panel econometric framework to test whether the thermal blanket mechanism operates with greater severity within formally recognized informal settlements. Delhi provides an ideal setting for this analysis owing to its extreme pre-monsoon thermal volatility and comprehensively documented informal settlement geography.

This paper asks: to what extent is the dual stressor of heat and pollution associated with a disproportionate reduction in nocturnal cooling within Delhi's informal settlements? We are explicit from the outset that, given an observational design without randomization, we cannot identify causal effects, and that our estimates measure associations conditional on the fixed effects described below. By quantifying this compound environmental association, we aim to provide empirical grounding for targeted, hyper-local climate resilience interventions rather than generalized city-wide mandates.

2. Study area, Datasets, and Methodology

2.1. Study Area

The analysis is focused on New Delhi. To ensure precise geospatial tracking and prevent spatial mismatch errors between various satellite instruments, we constructed a unified 3.5 km x 5.5 km fishnet grid, resulting in 78 distinct spatial cells across the territory. Within this geography, the study specifically examines highly vulnerable urban morphologies. To identify these informal settlements, we utilized official geographic coordinates for 675 Jhuggi Jhopri (JJ) clusters published by the Delhi Urban Shelter Improvement Board (DUSIB).

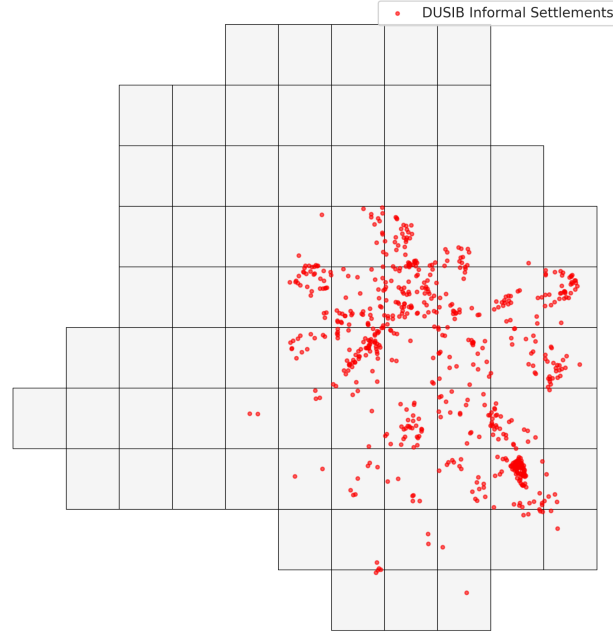


Fig 1: Spatial Grids and DUSIB settlements location

2.2. Datasets and Processing

Data acquisition and spatial harmonization were conducted via the Google Earth Engine (GEE) Python API. All longitudinal data were filtered strictly for the pre-monsoon summer months (April–June) across three years (2022 to 2024). The primary variables include:

- **Dependent Variable:** Daily Nocturnal Cooling Rate (ΔT), computed as daytime LST minus nighttime LST from MODIS Aqua (MYD11A1) at 1 km resolution.
- **Primary Exogenous Variable:** Aerosol Optical Depth (AOD) from Sentinel-5P (TROPOMI) at 1.1 km resolution, averaged daily per grid cell.
- **Robustness Variable:** Tropospheric NO_2 column density from Sentinel-5P (TROPOMI), used as a secondary pollution measure.

To merge the environmental data with our demographic focus, we performed a spatial intersection, flagging any of the 78 grid cells that contained at least one DUSIB slum coordinate for the subsequent panel analysis.

Variable	Source	Spatial Resolution	Temporal Coverage	Function in Model
Nocturnal Cooling (ΔT)	MODIS AQUA (MYD11A1)	1 km	Pre-Monsoon (Apr-Jun), 2022-2024	Dependent Variable (Y)
Aerosol Optical Depth (AOD)	Sentinel-5P (TROPOMI)	1.1 km	Pre-Monsoon (Apr-Jun), 2022-2024	Exogenous Variable (X_1)
Tropospheric NO ₂	Sentinel-5P (TROPOMI)	1.1 km	Pre-Monsoon (Apr-Jun), 2022-2024	Robustness Variable (X_1)
Informal Settlements	DUSIB (Delhi Govt)	Exact Coordinates (Point)	Time-Invariant	Interaction Binary (X_2)
Spatial Master Grid	FAO GAUL / Python	3.5 km x 5.5 km	Time-Invariant	Fixed Effects Entity Unit

Table 1: Data Sources

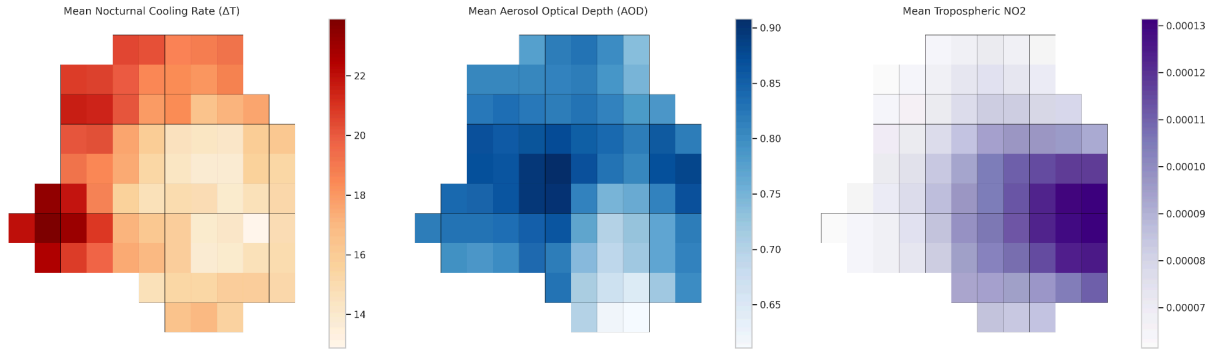


Figure 2: Spatial Distribution of Heat and Pollution Variables

2.3. Model Specification

To assess the association between localized pollution and nocturnal cooling, we employ a Two-Way Fixed Effects (TWFE) panel regression specified as:

$$\Delta T_{it} = \beta_1 \text{Pollutant}_{it} + \beta_2 (\text{Pollutant}_{it} \times \text{Informal}_i) + \alpha_i + \gamma_t + \varepsilon_{it}$$

where ΔT_{it} is the nocturnal cooling rate for grid cell i on day t , and Informal_i is a time-invariant binary indicator denoting the presence of at least one DUSIB settlement coordinate within the grid cell. The coefficient β_1 captures the average city-wide association between aerosol loading and reduced cooling. The interaction coefficient β_2 is the primary estimand, answering the question of whether the pollution-cooling association is stronger in informal settlement cells, above and beyond the city-wide effect. Entity fixed effects α_i absorb all unobserved, time-invariant characteristics of each grid cell (e.g., proximity to the Yamuna River, baseline land-cover composition, elevation). Time fixed effects γ_t absorb any shock that affects all grid cells on the same day (e.g., a regional heatwave, a city-wide wind shift). Standard errors are clustered at the grid-cell level to account for within-cell serial correlation.

2.4 Model Assumptions

The TWFE model relies on the assumption of strict exogeneity – that unobserved, time-varying factors do not systematically correlate with both pollution and cooling within a specific grid. While the fixed effects successfully absorb static morphology and daily macro-weather, hyper-local, time-varying changes (such as new localized construction during the three-year window) could introduce bias. Furthermore, we must assume no significant spatial spillovers (SUTVA), though atmospheric variables inevitably drift across boundaries.

Because observational data preclude a randomized control trial, definitive causal claims cannot be made. To stress-test the findings, we utilized Tropospheric NO_2 as a robustness check. If the thermal blanket effect were purely an artifact of regional aerosol transport, localized NO_2 (a heavy trace gas originating from immediate traffic and industrial combustion) should not exhibit a similar interactive penalty.

3. Results

3.1. The thermal blanket effect

Controlling for time-invariant grid characteristics (α_i) and daily city-wide shocks (γ_t), we find a negative association between aerosol loading and nocturnal cooling. A one-unit increase in AOD is associated with a 1.02°C reduction in the nocturnal cooling rate ($p < 0.001$). This is consistent with the thermal blanket mechanism described by Cao et al. (2016), whereby aerosols absorb and re-emit longwave radiation downward, reducing the rate at which the urban canopy cools overnight.

Critically, the interaction coefficient for grid cells containing DUSIB settlements is negative and statistically significant ($\beta_2 = -0.199$, $p < 0.001$). This indicates that, on high-AOD days, informal settlement cells are associated with approximately 0.2°C less cooling than formal-area cells, even after accounting for permanent geographic differences and shared daily weather. We interpret this as a robust correlation rather than a causal estimate for reasons discussed in Section 2.4.

	Parameter	Std Err	T-stat	P-value	Lower CI	Upper CI
const	18.562	0.1375	134.96	0.0000	18.292	18.832
aod_level	-1.0220	0.1249	-8.1814	0.0000	-1.2668	-0.7771
aod_x_informal	-0.1992	0.0514	-3.8773	0.0001	-0.2999	-0.0985

Table 2: TWFE Regression Results – Primary AOD Specification

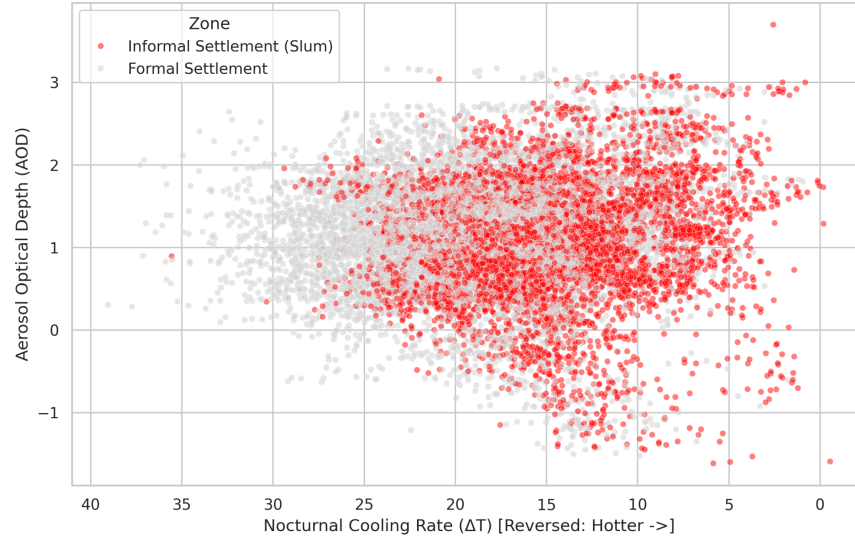


Figure 3: Interaction of Nocturnal Cooling vs. Aerosol Pollution

3.2. Robustness Check: NO₂ Combustion Specification

As a secondary specification and as a check for robustness, we substitute Tropospheric NO₂ as the pollution measure. NO₂ is a locally generated combustion trace gas originating from traffic and industrial activity, making it a useful complement to AOD, which captures broader regional aerosol transport. If the thermal blanket effect were purely an artifact of regional smog rather than local structural conditions, we would expect NO₂, which does not travel as far, to show a weaker or absent interaction with informal settlement status.

A one-unit increase in NO₂ is associated with a statistically significant reduction in nocturnal cooling ($\beta_1 = -5,127$, $p = 0.004$). The interaction term for informal settlements is negative in direction ($\beta_2 = -2,704$) and borderline significant ($p = 0.079$). The attenuation in significance relative to the AOD specification is expected: NO₂ is far more spatially heterogeneous than AOD, introducing greater measurement noise when averaged to the 1.1 km satellite resolution. The directional consistency across both pollutant measures suggests that informal settlement penalty is not an artifact specific to AOD.

	Parameter	Std Err	T-stat	P-value	Lower CI	Upper CI
const	17.949	0.1056	170.01	0.0000	17.742	18.156
no2_level	-5127.2	1760.7	-2.9121	0.0036	-8578.5	-1676.0
no2_x_informal	-2703.9	1537.1	-1.7591	0.0786	-5716.9	309.02

Table 3: TWFE Regression Results – NO₂ Robustness Specification

3.3. Threshold Sensitivity Analysis: Definition of “Informal”

Our baseline specification flags a grid cell as informal if it contains at least one DUSIB JJ coordinate. This is the most permissive threshold and likely overstates the informal character of some cells. To assess how sensitive our results are to this assumption, we re-estimate the primary AOD model under an additional alternative threshold:

- **Baseline:** Informal = 1 if the cell contains ≥ 1 JJ coordinate point.
- **Alternative:** Informal = 1 if the cell contains ≥ 5 JJ coordinate points.

Under the alternative threshold, the interaction coefficient is -0.231 ($p < 0.001$), slightly larger in magnitude than baseline, consistent with a cleaner signal as cells with marginal informal presence are reclassified to formal. The monotonic increase in magnitude with stricter thresholds suggests that cells with a higher density of registered JJ clusters experience a stronger penalty, which is consistent with the validity of the baseline finding.

4. Qualitative Deep-Dive: Ground Zero Settlements

Our composite vulnerability score ($z_{\text{aod}} - z_{\text{heat}}$) identified grid cell 7708,2863, which covers West Delhi’s Vikas Puri, as the highest-ranked cell in the entire 78-cell panel, with a composite penalty of 2.51. This single 3.5×5.5 km cell contains over ten officially registered DUSIB JJ clusters. The three settlements detailed below top that ranking list. Because they share the same grid cell, they also share identical model values, and this section explores why it’s important to examine the specific structural and social conditions through which each community experiences the same thermal trap differently.

4.1. Settlement #1: JJ Camp Indira Camp No-II, Vikas Puri

With 577 registered jhuggis on Delhi Development Authority (DDA) land, JJ Camp Indira Camp No-II is the largest single cluster in this grid cell. The “No-II” in its name tells the story that this is a second-generation settlement, one that expanded from an original encroachment rather than ever transitioning to formal housing, a pattern which is repeated across Delhi’s JJ landscape as informal settlements subdivide over decades [6,7]. The DDA’s land ownership can explain why the thermal penalty persists here specifically. Before any in-situ upgrading can proceed, the DDA must issue a No Objection Certificate [3]. Without one, permanent construction becomes grounds for eviction. Residents are therefore caught between

choosing either to improve the structure and risk losing the home, or leave the thermally vulnerable roofing in place. Decades later, the same galvanized iron and temporary sheeting that went up when the camp formed is still there absorbing heat during the day, and still releasing almost none of it overnight. The Centre for Policy Research states that JJ clusters on DDA land have “the least secure tenure of any settlement category in Delhi,” leaving them “most vulnerable to demolitions” [8]. Tenure insecurity and thermal vulnerability are not separate problems here, but rather the same one.

4.2. Settlement #2: Krishna Park, Vikas Puri Indira Camp No. 4

On paper, Krishna Park Indira Camp No. 4 looks much like its neighbors. What sets it apart is the use of pre-fabricated metal panels in the DUSIB Jan Suvidha Complex list [6, 7]. Prefabricated metal panel housing was first introduced in Delhi’s JJ clusters as a transitional upgrade, a step above improvised jhuggi construction. However, Metal panels carry almost no thermal mass and offer negligible insulation as they spike in temperature during the day and shed almost nothing after dark [9]. Mud brick buffers temperature swings, while Pucca concrete, for all its flaws, at least permits some radiative release overnight, while prefab panels do neither. Research on housing conditions across Delhi’s settlement types found that JJ cluster dwellings already suffer the worst indoor thermal environments in the city, with the prescription being to urgently improve “dwelling quality” [9]. Here, the state’s own upgrade program has locked in a material that makes these conditions worse. The prefab label is not a mark of improvement but rather a thermal liability written into official records.

4.3. Settlement #3: Shanker Garden B-Block, Vikas Puri

Both by household count and area, Shanker Garden B-Block is the largest cluster in the highest-ranked grid cell in our panel with 604 jhuggis and 9,235 square meters of land [6]. Each household is approximately 15 m², which is consistent with documented JJ cluster averages where mean floor areas run below 20 m² and household sizes average five or more occupants [10]. This results in having more than two people per room in structures that trap heat. The “B-Block” label signals decades of settlement, which is enough time to develop internal geography and establish community life. However, none of that longevity translates into thermal relief. Compact low-income neighborhoods in Delhi consistently show higher heat stress than less dense areas, precisely because density eliminates the ventilation gaps and tree cover that allow temperatures to fall at night [11]. At Shanker Garden, 604 households packed into under a hectare means no canopy and no convective pathway out. Body heat from hundreds of sleeping families compounds what the aerosol blanket is already doing above. The −0.199°C interaction penalty our model estimates does not get smaller as settlements get larger, indicating that this heat trap scales with density.

4.4. Synthesis

Across these three settlements, the satellite sees no difference as the data largely remains the same. The same AOD, ΔT , and composite score, all sit beneath the same aerosol blanket, in the same grid cell, under the same atmospheric conditions. Each community arrived at that shared exposure through a different history. A DDA tenure structure that makes roofing upgrades legally dangerous, a state-issued prefab classification that locks in thermally inert materials, and

a settlement density so extreme that no adaptation at the household level can compensate. The -0.199°C interaction penalty is not driven by one unusual cluster pulling the average down. It is rather the consistent outcome across communities with different histories and structures and relationships to the state, all ending up in the same thermal trap each night. What all of these settlements actually need are targeted interventions matched to each settlement's specific vulnerabilities.

5. Discussion

5.1. Interpretation

Our results provide robust evidence of an association between urban aerosol loading and reduced nocturnal cooling, and of an additional penalty for informal settlement grid cells. The directional consistency of the NO_2 specification further suggests the penalty is not merely a product of regional aerosol transport.

We are careful not to overstate what these findings establish. The TWFE model controls for important confounders, but it does not constitute a natural experiment. We cannot rule out that unobserved, time-varying factors— such as localized construction activity or changes in industrial land use within individual cells during 2022–2024 – partially explain the interaction term. As a result, our interpretation is that, conditional on permanent cell characteristics and common daily shocks, informal settlement cells exhibit a stronger negative association between pollution and cooling. Whether this reflects a causal mechanism or a residual confound is a question future work with higher-resolution or experimental designs could address.

5.2. Limitations

Observational design: TWFE controls for important confounders but cannot eliminate all unobserved time-varying factors.

Spatial resolution: MODIS (1 km) and Sentinel-5P (1.1 km) averages are aggregated to 3.5×5.5 km cells. Street-level micro-climate variation is smoothed out, and our estimates are limited to cell-level averages.

Binary informal flag: The baseline informal indicator is coarse, and finer-grained measures of settlement density or roofing material type would improve precision.

Potential spatial spillovers: Atmospheric variables drift across cell boundaries. Violation of SUTVA could bias the interaction coefficient in either direction.

Missing data: Cloud cover creates missing satellite retrievals that generate an unbalanced panel.

5.3. Stakeholder Map and Broader Implications

The implications of these findings require a multi-stakeholder approach to urban climate resilience. As outlined in our stakeholder map (Figure 4), the primary institutional actor is DUSIB. Current housing policies frequently treat slum rehabilitation strictly as an issue of land tenure or sanitation; our data argues it must concurrently be managed as an acute climate

emergency. Rather than relying solely on city-wide emissions targets, the compounded nature of this thermal penalty justifies hyper-local green infrastructure interventions, such as targeted cool-roofing subsidies and the establishment of ventilation corridors specifically within JJ clusters.

While this study is localized to Delhi, the methodological framework and resulting implications are relevant to other rapidly expanding cities in the Global South. As urban migration accelerates, understanding how informal morphology interacts with atmospheric physics is critical to ensuring that climate adaptation efforts reach the populations most heavily burdened by the built environment.



Figure 4: Stakeholder Map

6. Conclusion

This study provides evidence that the intersection of urban heat and air pollution is not experienced uniformly across the urban canopy. Using a Two-Way Fixed Effects panel model applied to 11,212 grid-day observations across three pre-monsoon seasons in New Delhi, we find a strong city-wide association between aerosol loading and reduced nocturnal cooling ($\beta_1 = -1.02^\circ\text{C}$ per AOD unit, $p < 0.001$), and an additional penalty for formally recognized informal settlement grid cells ($\beta_2 = -0.199^\circ\text{C}$, $p < 0.001$).

We emphasize that these are correlational findings conditional on the TWFE controls, not causal claims. The qualitative analysis of Vikas Puri settlements illustrates the plausible structural mechanisms – galvanized iron roofing, prefabricated metal panels, extreme residential density, and tenure constraints that prevent material upgrades – through which the thermal penalty is institutionally entrenched.

These findings suggest that climate resilience planning for informal settlements cannot rely on city-wide emissions targets alone. Because the cooling penalty correlates with urban morphology, it points toward interventions operating at the neighborhood level rather than the city level. Whether slum rehabilitation programs explicitly incorporate thermal vulnerability alongside land tenure and sanitation is ultimately a policy question, but the spatial specificity of the association identified here suggests that hyper-local targeting, informed by settlement-level vulnerability data, would be better positioned to reach the populations most affected.

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